
1. Introduction to JavaScript

JavaScript is a high-level, interpreted programming language.

It is primarily used for adding interactivity to web pages.

JavaScript can run in the browser (client-side) and on the server (Node.js).

2. Variables and Data Types

Declaring Variables

```
let name = "Alice";  
const age = 25;  
var isStudent = true;
```

let and const are block-scoped (introduced in ES6).

var is function-scoped (older syntax).

Data Types

String: "Hello"

Number: 10, 10.5

Boolean: true, false

Null: null

Undefined: undefined

Object: { key: "value" }

Array: [1, 2, 3]

3. Operators

Arithmetic: +, -, *, /, %

Assignment: =, +=, -=, etc.

Comparison: ==, ===, !=, !==, <, >, <=, >=

Logical: &&, ||, !

4. Control Structures

Conditional Statements

```
if (age >= 18) {  
  console.log("Adult");  
} else {  
  console.log("Minor");  
}
```

Switch Statement

```
switch (day) {  
  case "Monday":  
    console.log("Start of the week");  
    break;  
  default:  
    console.log("Another day");  
}
```

5. Loops

For Loop

```
for (let i = 0; i < 5; i++) {  
  console.log(i);  
}
```

While Loop

```
let i = 0;  
while (i < 5) {  
  console.log(i);  
  i++;  
}
```

For...of Loop (for arrays)

```
const fruits = ["apple", "banana", "cherry"];  
for (const fruit of fruits) {  
  console.log(fruit);  
}
```

6. Functions

Function Declaration

```
function greet(name) {  
  return "Hello " + name;  
}
```

Arrow Function (ES6)

```
const greet = (name) => "Hello " + name;
```

7. Arrays and Objects

Array Methods

```
const nums = [1, 2, 3];  
nums.push(4);    // Add  
nums.pop();      // Remove  
nums.includes(2); // Check
```

Objects

```
const user = {  
  name: "Alice",  
  age: 25,  
  greet() {  
    return `Hi, I'm ${this.name}`;  
  },  
};
```

8. DOM Manipulation (Browser)

```
document.getElementById("myDiv").innerText = "Updated!";  
document.querySelector(".btn").addEventListener("click", () => {  
  alert("Button clicked!");  
});
```

9. ES6+ Features

Destructuring: `const {name, age} = user;`

Spread Operator: `const copy = {...user};`

Template Literals: `'Hello, ${name}'`

Promises & Async/Await

```
async function fetchData() {  
  const res = await fetch("https://api.example.com");  
  const data = await res.json();  
  console.log(data);  
}
```